that exist be composed of a quantum world where nothing was real? “The great and compelling reason for refusing to regard the elementary particles as theoretical fictions,” wrote Smart, “is that unless something like what quantum mechanics tells us is true of some underlying reality, then [the fact] that the macroscopic laws are what they are… [is]too much of a coincidence to be believed.”

And there was another problem: the most straightforward way to make “measurement” fundamental to a theory was to be an operationalist—but operationalism was manifestly wrong. “Measurements are a subclass of physical interactions—no more or less than that,” wrote Putnam in 1965. “‘Measurement’ can never be an *undefined* term in a satisfactory physical theory, and measurements can never obey any ‘ultimate’ laws other than the laws ‘ultimately’ obeyed by *all* physical interactions.” Fixing the problem through a split between the micro-world of the quantum and the macro-world of everyday classical physics, as Bohr had insisted, didn’t help either. “Besides pushing the problem back to exactly the same problem in classical physics… the suggestion is completely unacceptable,” Putnam said. “We can hardly refer to one theory (classical physics) in the [foundations] of another [quantum physics] if the first theory is supposed to be incorrect and the second is designed to supersede it.… Quantum mechanics must, if correct, apply to systems of arbitrary size.… In particular, it must apply to macro-systems.” But if that’s the case, Putnam continued, then “what of macro-observables that are isolated for a long time, say, a system consisting of a rocket ship together with its contents out in interstellar space? We cannot seriously suppose that the rocket ship begins to exist only when it becomes once again observable from the earth or some other outside system.” For the Copenhagen interpretation, measurements were a serious problem. Smart agreed, with withering criticism of the idea that measurements must be described classically:

> Proponents of the Copenhagen interpretation of microphysics have been wedded to *classical* physics. They argue that since this is the physics of macroscopic instruments whereby we interpret our observations, this must remain stable whatever the advances in microphysics. That this is not so can be shown (as by Feyerabend) by posing one simple question: why *classical* physics? Why not, for example, Aristotelian physics, or even witchcraft, which was once just “scientific common sense”? Similarly, we must reject the view that there are